# Automatic text-to-speech system for Vietnamese use Artificial Intelligence

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**Abstract.** In the context of digital transformation and the rapid advancement of artificial intelligence, the demand for text-to-speech (TTS) systems has become increasingly critical across domains such as virtual assistants, education, automated call centers, and accessibility tools for the visually impaired. However, generating natural, context-aware, and prosodically appropriate speech remains a major challenge, especially for Vietnamese—a tonal language with complex pitch contours and regional diversity. This research proposes and develops a Vietnamese TTS system built on modern deep learning models to overcome the limitations of traditional methods. The system uses the latest architectures, including Tacotron, FastSpeech and VITS, combined with machine learning techniques to improve synthesis quality. Construction and evaluation are performed on a diverse Vietnamese dataset, ensuring adaptability to various real-world contexts. Results show that the system achieves natural, clear, highly flexible speech output with broad potential for deployment in digital platforms and intelligent services.
**Keywords:** Text-to-Speech (TTS), Artificial Intelligence (AI), Deep Learning, Natural Language Processing (NLP), Speech Synthesis.

## 1 Introduction

Driven by digital transformation and the advancement of artificial intelligence, the need to build text-to-speech (TTS) systems is becoming increasingly important in areas such as virtual assistants, education, automated call centers, and support for the visually impaired. However, the TTS problem extends beyond converting text to audio; it requires the synthesized voice to be natural, carry appropriate intonation, and fit the given context - a particularly difficult task for Vietnamese, a language with a complex tone system and considerable regional variation (Do, 2026). Traditional concatenative and formant-based TTS methods (Zen et al., 2019) and Hidden Markov Model (HMM)-based synthesis (Zen et al., 2019) exhibit limitations in flexibility, naturalness, and scalability. In response, modern deep learning models such as Tacotron (Wang et al., 2017), Tacotron 2 (Shen et al., 2018), FastSpeech (Ren) 2 F. Skrywer and S. Skrywer et al., 2019,2020), and VITS (Kim et al., 2021) have opened up more effective avenues by learning directly from data. Building on these advances, this study focuses on the research and development of a Vietnamese TTS system aimed at improving voice quality and practical applicability.

## 2 Related work

The Vietnamese TTS market features several prominent local platforms alongside international solutions. FPT Smart Cloud Text to Speech, developed by FPT Corporation, provides a high-quality Vietnamese TTS service through a commercial API (Vu et al., 2025; Nguyen, 2023). It applies deep learning architectures such as Tacotron (Wang et al., 2017) and Transformer variants (Vaswani et al., 2017) along with vocoders such as WaveNet (Oord et al., 2016) and HiFi-GAN (Kong et al., 2020) to produce voice and region control with diverse speed. The system is widely used in interactive voice response (IVR), audio books, virtual assistants and online education (Do, 2026). Viettel AI Text-to-Speech offers a SaaS-based solution optimized for Vietnamese (Vu et al., 2025; Nguyen, 2023), which uses advanced neural modelle wat on large-scale, Vietnamese-specific speech corpora are trained to ensure accurate 206 pronunciation, and expressive, natural pronunciation, ear and pronunciation. al., 2016; Kong et al., 2020). Zalo AI Text to Speech, a product of VNG Corporation, integrates deep NLP models such as &PhoBERT(Nguyen&Tuan, 2020) and BERT-style pretraining (Devlin et al., 2019) with synthesis architectures including Tacotron, FastSpeech, and Rens et al., VITS, 20 et al. 2019; Kim et al., 2021), which uses massive user data to support multiple speaking styles (Vu et al., 2025). Voice.vn specializes in speech synthesis and voice cloning (Vu et al., 2025) using end-to-end models such as VITS (Kim et al., 2021) and voice conversion techniques (Casanova et al., 2022; Chen et al., 2022), which enable customization for voice termination and customization for advertising. podcasting and gaming. Google AI Studio provides a cloud-based environment for TTS via the Gemini family and Google Cloud Text-to-Speech (Vaswani et al., 2017; Ren et al., 2019), with multilingual, natural and context-adaptive voice generation (She et al., 20182). EventLab offers TTS and voice cloning tools built on end-to-end TTS (Wang et al., 2017) and speaker embedding models (Chen et al., 2022; Kim et al., 2021; Casanova et al., 2022). Ear the generally move the mark van simple text reading systems to intelligent communication platforms, with strong trends in voice cloning (Jia et al., 2018) and multimodal AI.

The open source community has contributed a series of Vietnamese TTS models. VietTTS is a representative system built with encoder-decoder or transformer architectures (Wang et al., 2017; Vaswani et al., 2017; Ren et al., 2019), trained on carefully normalized Vietnamese data (Do, 2026) to handle tones, compound words, and sentence time structures, providing suitable near-time structures, providing real latency. e-learning and audiobook production (Nguyen, 2023; Vu et al., 2025). NeuTTS-Air focuses on lightweight, efficient inference, balancing voice quality and speed to run on CPUs or low-resource devices (Ren et al., 2019,2020; Kim et al., 2021), optimized for chatbots, IVR and IoT applications (Ping et al., 2017; Kalchbrenner et al., 2018). Kani-TTS is a large-scale model with about 370 million parameters, aiming for high expressiveness and professional sound quality (Kim et al., 2021; Casanova et al., 2022), which excels in audiobook and media production, but requires powerful GPUs for deployment (Ren et al., 2019; Kall et al., 2019 et al. 2018). VieNeu-TTS combines TTS with voice cloning, extracting speaker features via techniques such as AutoVC (Qian et al., 2019) and StarGAN-VC (Kameoka et al., 2018) and applying them to synthesized speech (Casanova et al., 2022; Jiabling et al., 2018), but also personalization. ethical and safety issues. ViXTTS serves as a demonstration project for voice cloning, illustrating the technology for rapid experimentation (Thinh, 2024; Kim et al., 2021). Valtec-TTS is a compact model optimized for extremely resource-constrained environments such as embedded devices and IoT (Ren et al., 2019,2020; Ping et al., 2017; Kalchbrenner et al., 2018), providing sufficient clarity for simple application notification and guidance. Collectively, these models illustrate the maturation of Vietnamese TTS, with specialized branches ranging from high-quality synthesis to lightweight deployment and voice personalization.

Data quality is the foundation for TTS. The VLSP Speech Corpus is a large, academically oriented dataset with multi-speaker recordings in controlled conditions (Ardila et al., 2020; Vu et al., 2025), richly annotated with phonetic and regional information (Yamagishi et al., 2019; Do, 2026). VIVOS is a smaller open-source corpus (approximately 10–20 hours) of clean-read speech (Ardila et al., 2020; Ito & Johnson, 2017), widely used for initial modeling despite limited emotional and real-world diversity (Do, 2026). Commercial datasets from FPT Corporation, Viettel Group and VNG Corporation are much larger and more detailed. FPT's dataset consists of hundreds to thousands of hours of fully annotated studio-quality recordings (Vu et al., 2025; Do, 2026; Oord et al., 2016; Kong et al., 2020). Viettel's data combines studio and real world surveys covering diverse accents (Vu et al., 2025; Yamagishi et al., 2019; Jia et al., 2018; Do, 2026). Zalo's data benefits from its broad user ecosystem, which includes contemporary language usage (Vu et al., &2025; Nguyen&Tuan, 2020; Devlin et al., 2019; Kim et al., 2021). In addition, VietTTS provides a basic open dataset with text-audio pairs sufficient for small-scale experiments (Nguyen, &2023; Ito&Johnson, 2017; Vu et al., 2025). In general, dataset scale, diversity, annotation depth and realism directly determine TTS performance.

## 3 System design

Early TTS systems relied on continuous and formant-based methods, which compose or mathematically model speech units, but suffer from inflexibility, high database costs, and unnatural prosody (Zen et al., 2019). Statistical parametric synthesis using Hidden Markov Models (HMM) improved flexibility, but still produced monotonous, less natural output (Zen et al., 2019). The advent of deep learning has brought end-to-end models such as Tacotron and Tacotron 2, which use encoder-decoder architectures with attention to direct map text to

The proposed system implements an emotional TTS pipeline: input text containing emotion tags is segmented according to emotion, each segment is synthesized via viXTTS, and the resulting waveforms are concatenated with silence gaps to produce the final audio file

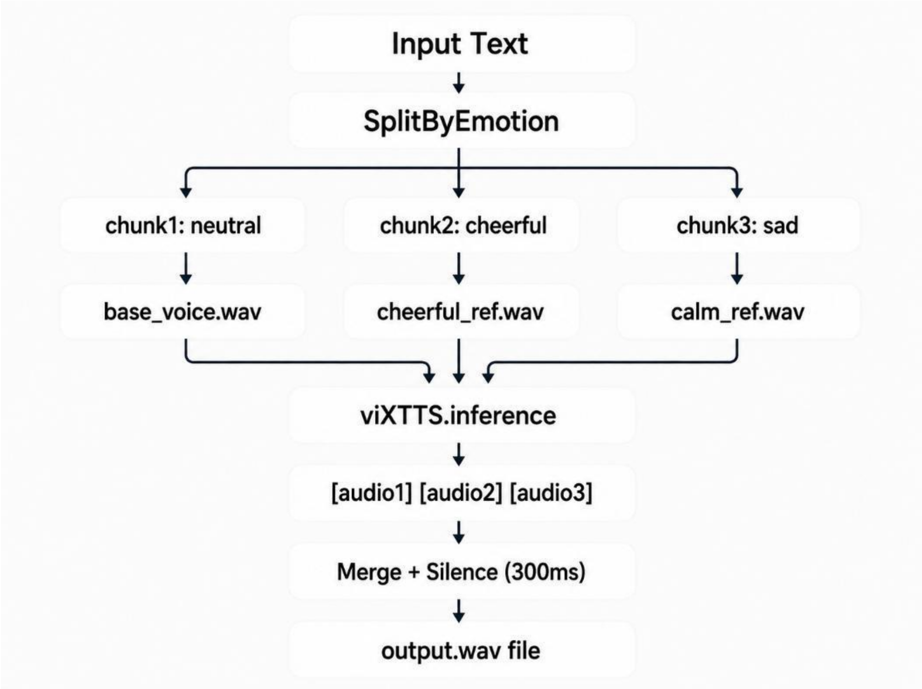


**Fig. 1.** *Workflow of the proposed emotion-based speech synthesis system*

In the first layer word text line for real scanned. When a line contains an emotion label in parentheses, the system converts it to an emotion map label using a Vietnamese key dictionary. Consecutive lines that share the same emotion are grouped into a chunk (text, emotion). For synthesis, the text is cleaned by removing markup, normalizing spaces, standardizing Vietnamese orthography. In the second layer, each part is passed to the viXTTS inference engine together with speaker marks extracted from a reference audio file. An emotion card assigns a reference file per emotion (eg neutral→base_voice.wav, cheerful → cheerful_ref.wav). To maintain consistent speaker identity, latent speaker embeddings are extracted once from a user-provided voice sample and reused across all chunks (Jia et al., 2018; Casanova et al., 2022). The synthesis produces a waveform for each

[redacted]

sad→calm_ref.wav, neutral→base voice.wav } ALGORITHM EmotionalTTS(text, ref_audio, output.wav): chunks ←split_for_emotion(text)
AS
pieces
muscle
is: ERROR (latent, embed) ←get_speaker_latent (ref_audio) waveforms ←[] FOR EACH (t, e)
IN
chunks: w ←viXTTS.inference(t, latent, embed) waveforms.append(w) output ←concatenate waveforms with 300 ms silence between
SAVE output →output.wav

[redacted]

repeat each line; if line '(' → close contains currentPart , detect_emotion(line) , begin new

[redacted]

[redacted]

each piece: clean_text (remove tags, normalize vietnamese) return pieces

## 4 Evaluation

Metrics To fully assess the text-to-speech and voice cloning system, two complementary deep-learning-based metrics are used: speaker similarity via Resemblyzer, and perceived speech quality via DNSMOS.

Resemblyzer is a neural network-based tool for speaker verification and voice similarity measurement (Mishra et al., 2023). At its core, it uses a speaker verification encoder trained with a common end-to-end loss (Wan et al., 2018). The operational principle is as follows: first, an input utterance—either the original reference voice or the synthesized cloned voice—is processed by a deep neural network that extracts a fixed-dimensional embedding vector (often referred to as a d-vector or speaker embedding). This vector is designed to capture the unique vocal identity of the speaker, including characteristics such as timbre, pitch contour, and speaking style, while being largely invariant to the linguistic content or the specific words spoken. In other words, regardless of what the person says, the embedding must remain consistent for the same speaker and distinctly different for different speakers (Wan et al., 2018).

Sodra die inbeddingsvektore vir beide die verwysingsoudio A (die oorspronklike spreker se stem) en die gesintetiseerde oudio B (die gekloonde stem) verkry is, word hul ooreenkoms gekwantifiseer deur die cosinus-ooreenkoms metriek te gebruik:

$$\text{CosineGelykvormigheid}(A, B) = \frac{A \cdot B}{\|A\|\|B\|} = \frac{\sum_{i=1}^{d} A_i B_i}{\sqrt{\sum_{i=1}^{d} A_i^2}\sqrt{\sum_{i=1}^{d} B_i^2}}$$

The cosine similarity measures the angle between the two vectors in the high-dimensional embedding space. A value close to 1 indicates that the vectors point in almost the same direction, meaning that the two voices are considered highly similar or almost identical in terms of speaker identity (Jia et al., 2018; Wan et al., 2018). Conversely, a value close to 0 or negative will represent different speakers. Resemblyzer therefore provides an objective, automatic and reproducible measure to evaluate how faithfully a voice cloning system preserves the original speaker's vocal characteristics.

DNSMOS (Deep Noise Suppression Mean Opinion Score) is a non-invasive, deep learning-based measure designed to estimate the perceptual quality of speech signals without requiring a clean reference audio signal (Reddy et al., 2021). That makes it

## 5 Experiments and results

**Training** dataset. The system was trained and evaluated using the viVoice corpus (Le, 2024), the first publicly available large-scale Vietnamese speech dataset designed for verbose text-to-speech. viVoice consists of over 1,000 hours of cleaned audio and accompanying transcripts sourced from 186 YouTube channels. All audio samples are provided at a 24 kHz sampling rate in WAV format, with background music and noise removed to ensure clean training data. The dataset covers a wide variety of speakers, accents (Northern, Central, Southern) and speaking styles, making it particularly suitable for training robust multi-speaker Vietnamese TTS models such as XTTS-v2. viVoice is released under the CC BY-NC-SA 4.0 license.

The preprocessing pipeline is designed to eliminate variability that can degrade speaker agreement. Steps included: (i) manual verification of transcript accuracy to ensure perfect alignment; (ii) text normalization using VietNormalizer (Do, 2026) to expand abbreviations, convert numbers to words and standardize punctuation; (iii) phonetic alignment with a forced alignment trained on Vietnamese, yielding phoneme-based framework; silence stripping at the beginning and end of utterances to remove leading/closing silence; and (v) loudness normalization to -23 LUFS to maintain consistent perceived volume across speakers. Crucially, for each speaker a high-quality speaker embedding was extracted using the Resemblyzer encoder and stored as a reference vector. Conditioning the generative model on these embeddings during training encourages the decoder to preserve speaker identity, which directly contributed to the high cosine similarity scores (0.90+) observed in the experiments.

**Experimental** setup. The evaluation was performed on a GIGABYTE G5 GD laptop, a gaming-oriented notebook whose specifications are listed in Table 1. In particular, the system has a dedicated NVIDIA GeForce RTX GPU (Ampere architecture, discrete) along with an integrated Intel UHD graphics adapter. However, the PyTorch environment used for derivations was a CPU-only build, meaning that all synthesis operations ran on the Intel Core i5-11400H CPU without GPU acceleration. This setup reflect 'n algemene deployment scenario where CUDA drivers or libraries are not installed, or where the system prioritizes compatibility over maximum speed. Table 1. System configuration of the test platform.

| Komponent | Specification |
|---|---|
| Model | GIGABYTE G5 GD |
| EVERYONE | Intel Core i5-11400H (8 cores, 12 threads, 2.70 GHz basis) |
| RAM | 16 GB DDR4 (15.8 GB usable) |
| GPU | Intel UHD Graphics (integrated) |
| Discrete GPU- | NVIDIA GeForce RTX (Amperes, Dedicated) |
| operating system | Windows 11 Home Single Language |

| | |
|---|---|
| Derivation Backend | PyTorch CPU only, 4 threads |

**Voice similarity and speech** quality. The system was evaluated by comparing original and synthesized (cloned) speech samples from four speakers. Each monster pair het exist out 'n reference.wav file en she kloon. Twee assessment methods were applied: Resemblyzer for voice similarity (Wan et al., 2018; Mishra et al., 2023), and DNSMOS for perceived sound quality (Reddy et al., 2021). Voting agreement results are presented in Table 2. All pairs reach a cosine similarity of approximately 0.90 (ranging from 0.9013 to 0.9346), confirming that the VITS model, despite using the CPU, preserves vocal timbre and speaker identity remarkably well (Jia et al., 2018).

Voting similarity results for the viXTTS model are presented in Table 2. All pairs reach a cosine similarity of approximately 0.90 (ranging from 0.9013 to 0.9346), confirming that the model preserves vocal timbre and speaker identity remarkably well (Jia et al., 2018). Speaker similarity for other engines was not measured in this evaluation run, as most of them do not natively support voice cloning or require a separate reference embedding extraction step. Table 2. Voice similarity between original and cloned speech.

| Lekker klank | Similarity |
|---|---|
| giong_goc af giong_clone | 0,9038 |
| Danh_1 en Danh_clone | 0,9019 |
| Khang1 and Khang_clone Ngoc1 | 0,9346 |
| en Ngoc_clone | 0,9013 |

Speech quality results from DNSMOS across all seven evaluated engines are summarized in Table 3. Four measures were recorded: OVRL (overall quality), SIG (speech quality), BAK (background noise intrusiveness), and the supplementary P808 score (an overall listening quality estimator aligned with ITU-T P.808). Scores are on the standard 1–5 mean opinion score scale, where higher values indicate better perceptual quality.

Table 3. DNSMOS evaluation of synthesized Vietnamese speech quality..

| Model | File audio | OVRL MOS | SEE MOSS | BAKE MOSS | P808 MOS |
|---|---|---|---|---|---|
| Proposed viXTTS-based Voice cloning system | giong_clone.wav | 3,38 | 3.60 | 4.14 | 4.03 |
| Piper Vietnamese TTS | Piper_Vietnamese_TTS | 3.00 | 3,40 | 3,81 | 3.15 |
| Valtec Vietnamese TTS v2 | Valtec Vietnamese TTS_ver2.wav | 3.07 | 3,34 | 4.04 | 3,95 |
| Vietnamese TTS | VietnameseTTS.wav | 2.07 | 2,75 | 2,86 | 2,73 |
| Valtec Vietnamese TTS | Valtec_Vietnamese.wav | 3,37 | 3,58 | 4.16 | 4.21 |
| VietTTS | VietTTS.wav | 3.05 | 3,25 | 4.15 | 3,64 |
| VieNeu-TTS | VietNeu_TTS.wav | 3.31 | 3,54 | 4.16 | 3,68 |

Among the compared systems, Valtec_Vietnamese.wav also performs competitively, with an OVRL MOS of 3.37 and a SIG MOS of 3.58, which is very close to the proposed system. In addition, this model achieves the highest P808 MOS score of 4.21, indicating that listeners may perceive its overall sound quality favorably. However, the proposed giong_clone.wav still slightly outperforms it in the two main DNSMOS indicators, OVRL and SIG. This indicates that the proposed model has a small but measurable advantage in terms of overall speech quality and signal clarity.

The VietNeu_TTS.wav model also achieves strong results, with an OVRL MOS of 3.31, SIG MOS of 3.54, BAK MOS of 4.16 and P808 MOS of 3.68. This scores place it in the high performing groep, veral in terme van background noise reduction. Its BAK score is among the highest in the comparison, showing that the generated sound is clean and stable. Although its overall and signal quality scores are slightly lower than those of giong_clone.wav and Valtec_Vietnamese.wav, it remains a reliable model for Vietnamese speech synthesis.

The middle performing group includes Valtec Vietnamese TTS_ver2.wav, VietTTS.wav and Piper_Vietnamese_TTS. These models achieve OVRL MOS scores ranging from 3.00 to 3.07, indicating acceptable but not outstanding synthesized speech quality. Their SIG MOS scores, which range from 3.25 to 3.40, indicate that the speech signal is quite clear, but still less natural or less detailed than the top performing systems. In particular, VietTTS.wav achieves a high BAK MOS score of 4.15, which shows that the audio output is relatively clean. However, the lower OVRL and SIG scores indicate that background beauty alone is not sufficient to ensure high overall speech quality.

The lowest performing system is VietnameseTTS.wav, which achieves an OVRL MOS of 2.07, SIG MOS of 2.75, BAK MOS of 2.86 and P808 MOS of 2.73. These scores are significantly lower than those of the other models. In particular, the low OVRL and SIG scores indicate that the synthesized speech may contain noticeable artifacts, reduced naturalness, or unclear pronunciation. The low BAK score also indicates that the audio may include more background interference or noise compared to the other systems.

A general trend can be observed across the stronger models: most systems achieve high BAK MOS scores above 4.0, meaning that modern Vietnamese TTS models are generally effective at producing clean audio with limited background noise. However, differences in OVRL and SIG scores show that clean sound is not always guarantees natural and understandable speech. Speech naturalness, clarity and signal quality remain important factors in distinguishing the best performing models.

In summary, the proposed viXTTS-based voice cloning model giong_clone.wav achieves the best overall performance in terms of OVRL and SIG MOS, what it the strongest system in this comparison maak. Valtec_Vietnamese.wav and VietNeu_TTS.wav also demonstrate competing performance and can as strong alternatives considered word. Meanwhile toon VietnameseTTS.wav the worst results and may require further optimization to improve speech naturalness, clarity and background noise control.

**Real time achievement- and latency** analysis: : The most critical performance measure of the current prototype is the end-to-end synthesis latency. For a typical input text of 750 characters (equivalent to about 50 seconds of generated audio at a standard Vietnamese speaking rate of about 900 characters per minute), the system takes about 2 minutes and 50 seconds (170 seconds) to produce the final WAV file. This yields a Real-Time Factor (RTF) of around 3.4, well above the real-time threshold of 1.0. However, the latency is not fixed; it varies measurably depending on the linguistic complexity of the input text. Two factors dominate the additional processing fee:

- Emosie-etikette. Elke emosionele grens dwing 'n nuwe sintese-stuk af met 'n aparte viXTTS-afleidingsoproep, wat lei tot veelvuldige konteksskakelaar en modellaai bokoste. Meer merkers verhoog die aantal stukke en die kumulatiewe pouse-invoegtyd.
- Engelse woorde en nie-Viëtnamese segmente. Dit vereis ekstra grafeem-tot-foneemstappe binne die teksnormaliseringslaag (Do, 2026), en kan aandagfoute veroorsaak wat die dekodeerder dwing om herbelyning te probeer, wat bydra tot die totale looptyd.

The observed latency results from running the full VITS pipeline on the CPU without using the available RTX GPU. The CPU-only PyTorch build cannot exploit the massive parallelism of the GPU's tensor cores, leaving matrix multiplications and convolutions to be computed sequentially. Despite this, the system remains acceptable for near-line content creation tasks such as podcast dubbing, educational narration or voicemail generation – applications where a few minutes of rendering time is tolerated in exchange for high-quality, emotionally expressive output.

The machine is critically equipped with an NVIDIA GeForce RTX GPU, which is fully capable of accelerating the derivation. Moving to a CUDA-enabled PyTorch installation will reduce the RTF to well below 0.1 (based on typical GPU benchmarks for VITS), enabling real-time stream synthesis. In addition, model quantization to FP16 or INT8, and switching to optimized inference engines such as ONNX Runtime with OpenVINO, can further halve latency while the high agreements en MOS scores behoue bly.

## 6 Conclusion

In this research, a Vietnamese text-to-speech system based on artificial intelligence and deep learning has been thoroughly designed, implemented and evaluated.

The system

The results achieved not only confirm the effectiveness of deep learning in Vietnamese speech synthesis, but also highlight the system's broad applicability in practice, especially in virtual assistants, education, automated call centers and aids for the visually impaired. Future work may focus on expanding and enriching the speech dataset, improving local naturalness, optimizing models for resource-constrained devices, and integrating the system into real-world platforms to improve feasibility and usability.

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